Probing Ultrafast Photocurrents

Investigating hot-carrier dynamics in Silicon single crystals using photocurrent spectroscopy

B.J. Spijkerman

October 30, 2026

Motivation & Background

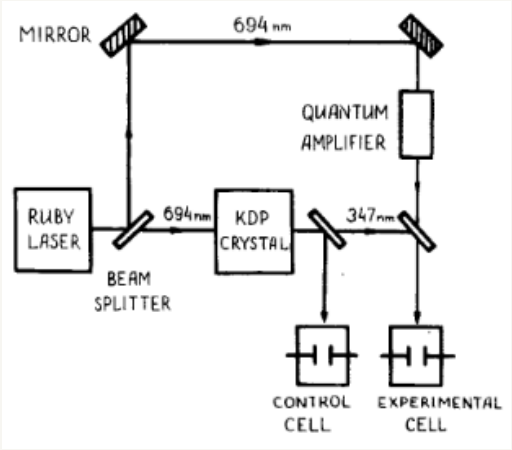
Utrecht University

- **Shockley-Quaser:** Thermodynamical limiting efficiency for solar panels.
- **Hot carrier-dynamics:** Very little known about hot state conductivity.
- **Photocurrent spectroscopy:** Measuring carrier-dynamics with pico-second resolution.

Figure:

Photo- current/ conductivity Spectroscopy

- Optical pump electrical probe
- Electrical detection
- Advantages over TA methods



L.V Lukin, *et all*, Chemical Physics Letters, Volume 81, Issue 3, 1981.

Experimental Methods

Utrecht University

Conductivity changes

The material responds to illumination by changing its conductivity:

$$\Delta\sigma = \Delta\mu\Delta N \tag{1}$$

Key Properties

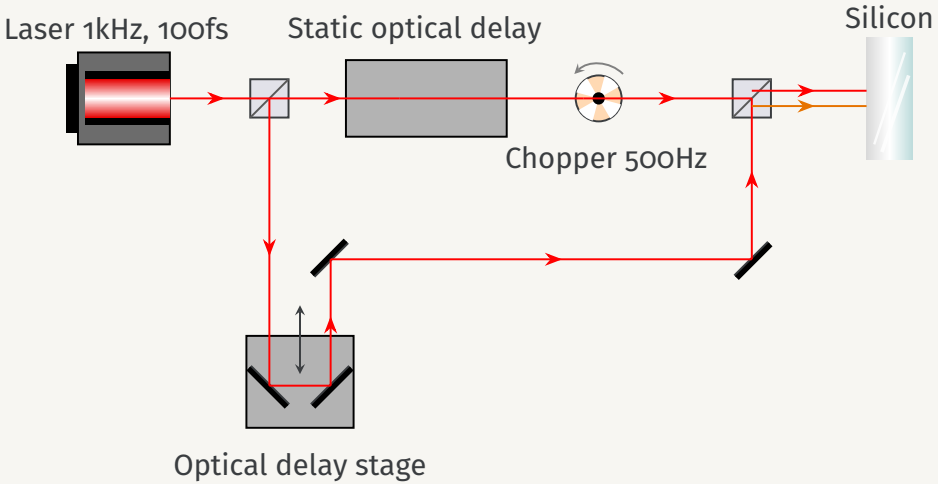
- $\Delta\sigma = \Delta\sigma_1 + \Delta\sigma_2 + C(\tau)$
- μ is related to carrier-dynamics.
- N is related to carrier-population.

Important Consideration

The different states have their own carrier mobility μ_i.

$$\Delta\sigma = \sum_i \mu_i N_i - \sigma_0 \tag{2}$$

Experimental Setup



Silicon

- Band structure
- Indirect bandgap
- Hot carriers

images on silicon

Carrier Two-Temperature Model

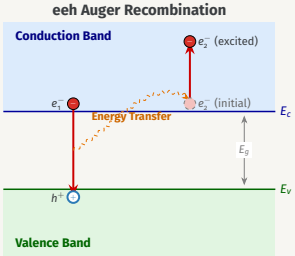
- Carrier population
- Carrier temperature
- Lattice temperature
- Conductivity

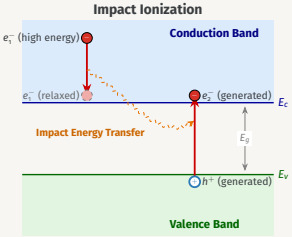
reference to figure

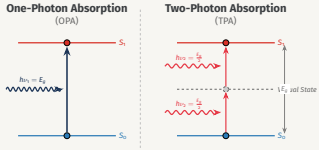
Carrier Two-Temperature Model

Carrier Population Model

$$\frac{\partial n}{\partial t} = -\gamma n^3 + \eta n + \frac{\alpha}{\hbar\omega_0} I + \frac{\beta}{\hbar\omega_0} I^2 \tag{3}$$







Carrier Two-Temperature Model

Utrecht University

Carrier Temperature Model

Carrier temperature model

$$C_c \frac{\partial T_c}{\partial t} = -K\left(T_C - T_l\right) + \left(\alpha + \beta I + \Theta n\right) I - \frac{\partial n}{\partial t}\left(E_g + 3k_B T_c\right) - n\left(\frac{\partial E_g}{\partial n}\frac{\partial n}{\partial t} + \frac{\partial E_g}{\partial T_c}\frac{\partial T_c}{\partial t}\right) \quad (4)$$

Carrier Two-Temperature Model

Utrecht University

Lattice Temperature Model

Lattice temperature model

$$C_l \frac{\partial T_l}{\partial t} = K(T_c - T_l) \tag{5}$$

Carrier Two-Temperature Model

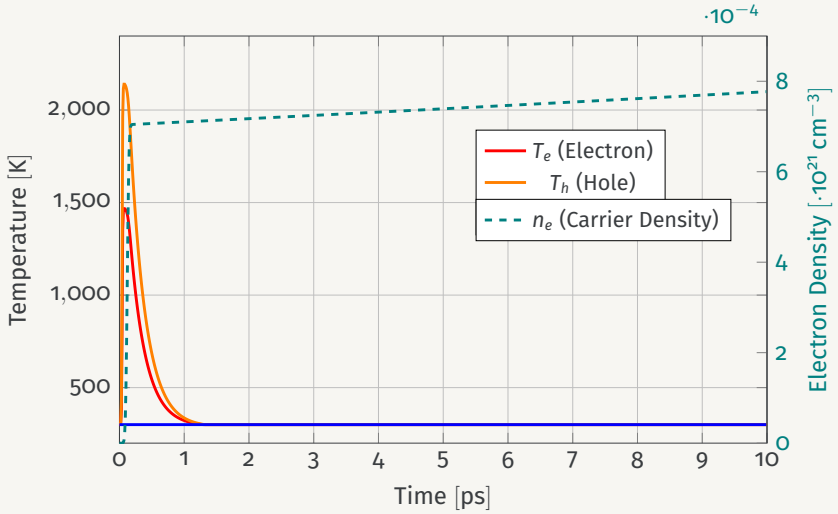
Utrecht University

Drude Model

Drude conductivity

$$\sigma = \frac{\sigma_0}{\nu\left(T_l\right)} \tag{6}$$

Carrier Two-Temperature Model

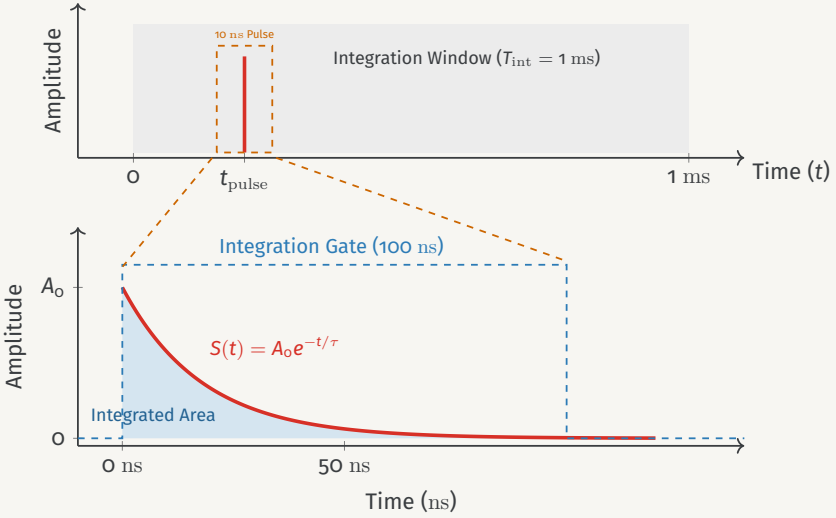


Electrical Detection

Utrecht University

- 500 Hz modulation
- 1 kHz repetition rate
- Lock-in amplifier SR-830
- Box car averager

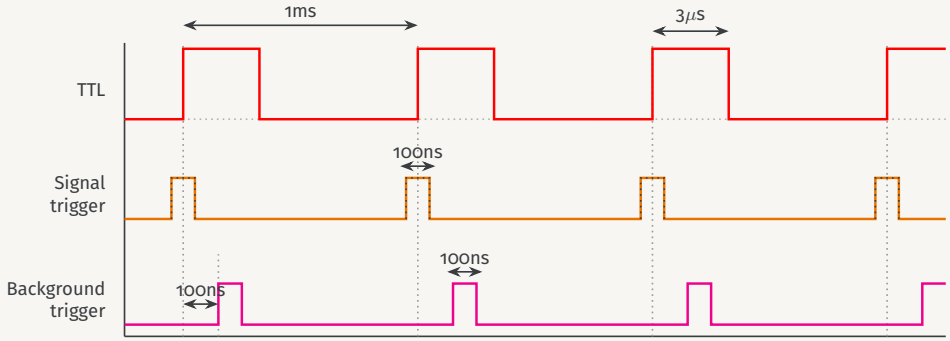
Electrical Detection



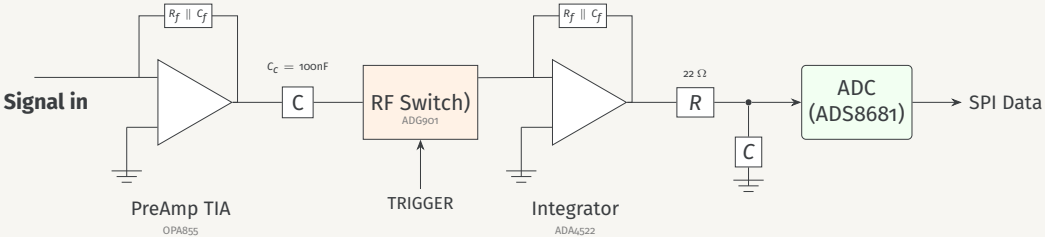
Electrical Detection

Box car

$$A[k] = \frac{1}{N} \sum_{i=0}^{N-1} \left(Signal\,[i] - Background\,[i] \right). \tag{7}$$

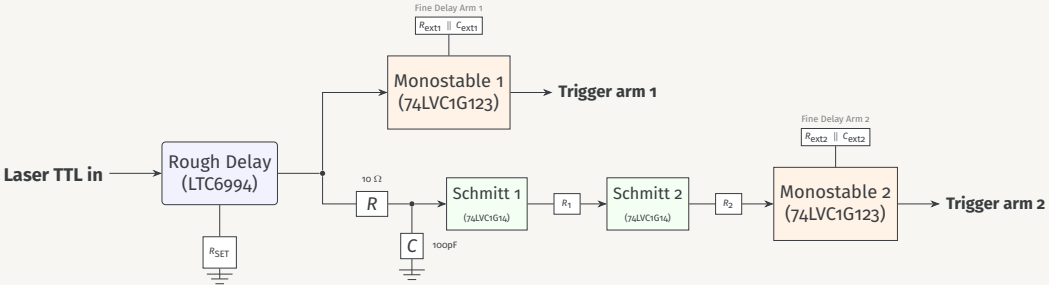


Electrical Detection



Component	Bandwidth [GHz]	Noise [$V_{rms}/\sqrt{Hz}$]
Pre-amp	8	0.342
Gated integrator	86.15	0.281
ADC	88.38	0.234

Electrical Detection



Trigger stage	Imparted delay	Phase jitter [ppm/Hz]	Pulse duration
Rough delay	1 ms	0.342	1 µs
Fine delay	100 ns	0.281	1 µs
Trigger generation	500 ps	0.234	100 ns

Electrical Detection

Utrecht University

SNR formula

The formula for signal-to-noise ratio assuming *abc*

$$SNR = somecomplicatedformula \tag{8}$$

Figure:

Different Photo- current/ conductivity Techniques

Utrecht University

- Correlation spectroscopy
- Vibrationally constrolled spectroscopy
- Near field
- ...?